DAY 12

Modules & Packages

INTERMEDIATE PHASE · DAYS 7–16

Session Length	Course Progress	PDF + .py + Problems
2 hrs	Day 12 / 30	3 deliverables

Import statements, the __**main**__ guard, installing packages with pip/uv, and five essential standard library modules: os, sys, math, datetime, pathlib.

30-Day Python Programming Course
Instructor Session Guide

1 · Session Overview

Goal: Students learn how to organise and reuse code across files using Python's import system, understand why the __**main**__ guard exists, learn how to install third-party packages, and become comfortable with five standard library modules they will use constantly for the rest of the course.

Outcome: By the end of this session, students can write multi-file Python projects, correctly guard demo/test code with __main__, and confidently reach for os, sys, math, datetime, and pathlib when solving real problems.

Session Timeline

Time	Activity	Details
0:00 – 0:15	Warm-up & Day 11 recap	Quick review of inheritance & dunder methods
0:15 – 0:40	Import statements & aliasing	import, from...import, as, avoid import *
0:40 – 0:55	Installing packages	pip install vs uv add, pyproject.toml
0:55 – 1:05	Break	—
1:05 – 1:30	The __main__ guard	Why it exists, importable vs runnable files
1:30 – 1:50	Standard library tour	os, sys, math, datetime, pathlib
1:50 – 2:00	Wrap-up & practice problems	Recap + assign day12 practice set

TIP

Keep the import-star warning concrete: show a real namespace collision live.
Have students run each demo file with `python filename.py` themselves.

2 · Prerequisites & Preparation

Requirement	From	Notes
Python 3.12 + uv installed	Day 1	Verify with `python --version`
Comfortable with functions	Day 5	Needed for guard pattern demos
try/except basics	Day 9	Used in mixed challenge problems
Classes & objects	Day 10–11	Used in Subscription class challenge

Instructor Pre-Session Checklist

■ Confirm all Day 10–11 .py files run cleanly on instructor machine
■ Open a terminal and pre-test `pip install requests` and `uv add requests` (don't actually leave it installed if not needed)
■ Create an empty 'weather_logs' and 'reports' folder structure for the pathlib demo, or let students see it get created live
■ Have day12_01, day12_02, day12_03 .py files open in VS Code tabs, ready to run live

3 · Step-by-Step Instruction Guide

STEP 1 — Import Statements & Aliasing (25 min)

> ***What to Say:*** *"Every .py file we've written so far has lived in isolation. Today we learn how to pull code IN from other files — both ones Python ships with, and ones other people have published."*

```
# Three import styles
import math
from math import sqrt, floor
import datetime as dt

print(math.pi)
# Output: 3.141592653589793
print(sqrt(49))
# Output: 7.0
```

Common Error	Cause	Fix
ModuleNotFoundError	Package not installed in this venv	Run pip install / uv add, check active venv
NameError: name 'sqrt' is not defined	Used `import math` but called sqrt() directly	Use math.sqrt() OR `from math import sqrt`
ImportError: cannot import name	Typo in the name being imported	Check exact spelling/case of the function/class

DONE WHEN
Student successfully imports math, uses both styles, and explains in one sentence why `import *` is risky.

STEP 2 — Installing Packages (15 min)

What to Say: *"math and os come free with Python. But the moment you want something like `requests` for web APIs (Day 23), you have to install it into your virtual environment first."*

Platform	pip (classic)	uv (modern, Day 1 tool)
macOS / Linux	pip install requests	uv add requests
Windows	pip install requests	uv add requests

TIP

uv automatically records the dependency in pyproject.toml —
this matters a lot for the Day 26-30 capstone project.

STEP 3 — The __main__ Guard (25 min)

What to Say: *"Python sets a hidden variable called __name__ in every file. If you run the file directly, it equals the string '__main__'. If another file imports it, it equals the file's own name. This is how we keep demo code from running by accident."*

```
def calculate_gst(amount, rate=0.18):
    return round(amount * rate, 2)

if __name__ == "__main__":
    print(calculate_gst(500))
# Output (only when run directly): 90.0
```

Common Error	Cause	Fix
Demo code runs on import	No __main__ guard around test/print code	Wrap demo calls in if __name__ == '__main__':
IndentationError after guard	Forgot to indent code inside the if block	Indent all guarded code by 4 spaces
Guard never triggers	Typo: '__main__' written without underscores	Must be exactly __name__ == "__main__"

DONE WHEN

Student can explain, in their own words, what __name__ equals when a file is run directly vs imported.

STEP 4 — Standard Library Tour (20 min)

What to Say: *"These five modules — os, sys, math, datetime, and pathlib — will show up in almost every project you ever write in Python, including our Task Manager capstone in week five."*

```
import os, sys, math, datetime
from pathlib import Path

print(os.getcwd())
print(sys.platform)
print(math.gcd(48, 180))
# Output: 12
print(datetime.date.today())
print(Path.cwd() / 'reports' / 'data.csv')
```

Module	Most-Used For
os	Working directory, environment variables, listing files
sys	Command-line args, Python version, exiting a program
math	Rounding, powers, logs, gcd, float comparisons
datetime	Dates, times, durations, formatting with strftime
pathlib	Modern, cross-platform file path building

DONE WHEN

Student builds a Path using the / operator and explains the difference between os.path strings and pathlib.Path objects.

4 · Session Wrap-Up

Student Checklist

- ■ Can write import, from...import, and aliased imports correctly
- ■ Knows the difference between pip install and uv add
- ■ Can explain what __name__ == '__main__' checks for and why
- ■ Has used at least os, math, datetime, and pathlib in working code
- ■ Completed day12_01, day12_02, day12_03 concept files without errors

Key Concepts Recap

Modules let us reuse code across files. **import** brings in a whole module; **from...import** brings in specific names; **as** creates an alias. Third-party packages need installing first (pip or uv). The **__main__** guard keeps demo/test code from running on import. The standard library modules os, sys, math, datetime, and pathlib cover file systems, system info, math, and dates.

TIP

Day 13 — Iterators & Generators: __iter__/__next__ protocol, the yield keyword, generator expressions, and itertools.

Troubleshooting Notes

If a student's package install seems to 'disappear' between sessions, check that they are activating the SAME virtual environment created on Day 1 — a very common issue is creating a new venv accidentally or running pip outside the venv.

5 · Quick Reference Card

Daily Workflow Commands

```
uv venv                    # create virtual environment (Day 1)
source .venv/bin/activate  # activate venv (macOS/Linux)
.venv\Scripts\activate     # activate venv (Windows)
uv add <package>           # install a package
python filename.py         # run a script
```

VS Code Shortcuts

Shortcut	Action
Ctrl/Cmd + `	Open integrated terminal
F5	Run/debug current file
Ctrl/Cmd + Shift + P	Open command palette

Tool Summary

Tool	Purpose
pip / uv	Install third-party packages
__name__	Detect direct run vs import
pathlib.Path	Cross-platform file paths